

Bibek Bhatta

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Summary

Mechanical Engineering undergraduate focused on aerospace autonomy and robotics. Designs full mechatronic systems spanning CAD and FDM hardware, embedded computer vision, and autonomous control, with hands-on UAV, drivetrain, and perception experience.

Technical Skills & Certifications

- CAD / CAE: SolidWorks, Onshape, FreeCAD; SolidWorks Flow Simulation (CFD), FEA, GD&T, tolerance stack-up; gear and drivetrain design
- Manufacturing: FDM 3D printing (PLA / PLA+ / ABS), DFM, design for thermal management, 3030 extrusion fixtures
- Mechatronics: EDF and tilt-mechanism design, NEMA steppers, lead-screw linear motion, cycloidal and planetary gearboxes, Arduino, GRBL, Altium PCB design
- Programming / Perception: Python, PyTorch, OpenCV (ArUco), Scikit-learn, NumPy, FastAPI
- Embedded / Autonomy: NVIDIA Jetson (Orin Nano), Pixhawk / PX4, GPS waypoint navigation, geofencing and fail safes, A* path planning, dataset annotation

Projects

Autonomous VTOL Medical-Delivery UAV

Personal Research via MSU Robotics Club / Starkville, MS

2026 - Present

- Designing a 1 m-wingspan tri-EDF tiltrotor VTOL (two wing-mounted and one tail-mounted ducted fan, all tilting in a V-22 Osprey-style configuration) for autonomous medical-supply delivery, a concept developed since high school and brought into full CAD and prototyping at MSU; currently lifting 800g against a 3 to 5 kg design target.
- Built a fully 3D-printed airframe and modular cargo bay for rapid prototype iteration, powered by LiPo 5200 mAh 7.2 V (2S).
- Sized wing geometry, propulsion, and stability margins using hand calculations and spreadsheet models, validated with SolidWorks Flow Simulation (CFD), iterating the airframe to balance range, payload, and stability.
- Integrated a Holybro Pixhawk Standard Baseboard V2A running PX4, implementing GPS waypoint-following with geofence and altitude failsafe that auto-recover when the aircraft exits its designated zone or ceiling.
- Prototyping autonomous routing with A* path planning and evaluating voxel-based methods for 3D obstacle-aware navigation; road mapping a migration from the COTS controller to a custom flight computer for full control of the autonomy and sensor-fusion stack.

Experience

Design Lead

Robotics Club, Mississippi State University / Starkville, MS

January 2026 - Present

- Leading mechanical design for the club's entry in NASA's Fly Foundational Robots (FFR) ISAM mission, developing a prototype robotic grabber and self-assembling structural modules that autonomously join into a functional structure for an in-orbit-assembly demonstration.
- Designed a standardized, 3D-printable library of mechatronic subsystems (cycloidal and planetary gearboxes, lead-screw linear stages on NEMA steppers, and matching motor controllers) so any club team can print and prototype drivetrains to a common standard instead of sourcing or reverse-engineering COTS parts.

Undergraduate Researcher

Department of Sustainable Bioproducts, Mississippi State University / Starkville, MS

February 2026 – Present

- Designed a multi-illumination imaging rig to grade Gibson guitar tonewood (2A / 3A / 4A) for a PhD-led CNN quality-control model: a FLIR Blackfly camera with dual synchronized LEDs in a controlled dark enclosure, capturing three exposures per sample (each light independently, then combined) to characterize the specular figure of high-grade wood; fixtures 3D-printed in PLA+.
- Wrote Python capture software that triggers synchronized lighting and acquisition from a single keypress and assembled a labeled dataset of roughly 1,200 samples (about 3,600 images) across three grades.
- Designing a handheld field device integrating the lighting, camera, and trained model for in-the-wild grade testing by Gibson wood suppliers (July 2026).
- Designed an automated 4-camera gantry spanning roughly 2.4 m of travel for cuboidal-log imaging: a 3030-aluminum square fixture with four independently positionable FLIR Blackfly cameras plus whole-gantry travel, driven by NEMA 17 steppers through planetary and cycloidal (27:1) reducers sized for ample torque margin, with lead-screw motion on all axes, controlled by an Arduino running GRBL.

- Provisioned an NVIDIA Jetson Orin Nano (Linux and ML toolchain) for edge wood-chip moisture detection and annotated a 7,000-image training dataset; designed an active-cooled, dust-resistant enclosure (heatsink and fan) to hold the board below 50 deg C in-mill.

Undergraduate Researcher

Department of Plant and Soil Sciences, Mississippi State University / Starkville, MS

October 2025 – February 2026

- Designed and 3D-printed an automated vibratory seed-feeder that singulates seeds and streams them into a camera's field of view for hands-free, repeatable imaging.
- Built an ArUco-calibrated vision station using marker-based homography to correct camera perspective distortion and constrain the region of interest, feeding an ML model that counts seeds and extracts per-seed metrics (area, length, width); processed roughly 200 seeds per 5-second capture with no fixed object-count limit, populating a research seed-trait database.
- Validated the system on sunflower and soybean samples and delivered the imaging enclosure and pipeline to the research team.

Education

Bachelor of Science in Mechanical Engineering

December 2028

Mississippi State University • Starkville, Mississippi • Junior

- **Colvard Future Leader** Scholarship | **Freshman Academic Excellence** Scholarship | **Presidents List** Summer & Fall 2025 | **Dean's List** Spring 2026

Cambridge International Advanced Levels

August 2023

St. Xavier's College • Maitighar, Kathmandu